

Pneumonia Image Detection with Federated Learning: Comparative Insights and Hybrid Approaches

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Abstract: Pneumonia remains a significant global health challenge, particularly in low-resource settings where timely and accurate diagnosis is often hindered by limited medical expertise and uneven data distribution. Conventional deep learning approaches for pneumonia image detection rely heavily on centralized datasets, raising privacy concerns and limiting scalability across institutions. This paper addresses these challenges by advancing pneumonia image detection through federated learning (FL), enabling collaborative model training without sharing sensitive patient data. The objective is to compare FL with traditional centralized learning and develop a hybrid approach that integrates the strengths of both paradigms. The proposed framework employs convolutional neural networks (CNNs) as the backbone, trained across multiple simulated hospital nodes using the Federated Averaging (FedAvg) algorithm, with adaptive weighting for heterogeneous data distributions. Additionally, a hybrid strategy is implemented, combining FL-trained global models with localized fine-tuning to capture site-specific imaging characteristics. Comparative experiments were conducted on publicly available chest X-ray datasets partitioned to mimic real-world data silos. The study finding demonstrate that the hybrid FL approach achieved an accuracy of 94.6%, outperforming standalone FL (92.8%) and centralized baselines in generalization across diverse test sets. These findings highlight the potential of FL-enhanced hybrid strategies to deliver accurate, privacy-preserving pneumonia detection models, supporting scalable deployment across healthcare networks and resource-limited regions.

Keywords: Federated learning, pneumonia detection, chest X-ray, privacy-preserving AI, hybrid approach, medical imaging

I. INTRODUCTION

The world health impact of pneumonia is still very high, especially for susceptible groups like children under five and the old. Pneumonia is a life-threatening lung illness. Although medical tests have come a long way, it is still hard to find pneumonia early and correctly. This is especially true in places with few resources, like those that don't have easy access to radiologists or good healthcare facilities. Chest x-rays, which are the main way to diagnose pneumonia, need to be interpreted by a specialist, and the way they do this can vary a lot from place to place. In this situation, artificial intelligence (AI), especially deep learning (DL) models, has shown a lot of promise in automatically finding pneumonia in chest X-rays. This could make diagnosis faster, more accurate, and easier for more people to get. Usually, traditional deep learning methods depend on centralised training with big datasets

that have been labelled. But these kinds of methods make people worry about data privacy, ownership, and control, especially in the medical field where private patient data is involved. Also, it's not always possible to combine data from different hospitals or schools because of rules like HIPAA or GDPR. There is also variety because imaging methods, demographics, and disease rates vary from institution to institution. This makes it hard for a single centralised model to work well in a lot of different situations. Federated Learning (FL) looks like a good way to get around these problems because it lets institutions that aren't centralised work together to train models without sending real patient data. FL protects privacy by keeping data local and only sharing model changes [1]. This lets a global model be made that uses data from a variety of sources. But FL comes with its own problems, such as extra communication, problems with model convergence, and slower performance because clients' data is not spread out exactly the same way (non-IID).

A new way to find pneumonia is suggested in this study. It uses a mixed method that combines Federated Learning with centralised learning techniques. The study aims to look at how well traditional centralised models, pure FL models using the Federated Averaging (FedAvg) method, and a mixed approach that combines FL-trained global models with localised fine-tuning to better catch site-specific differences in chest X-ray data work. Convolutional neural networks (CNNs) are used as the main structure in the suggested method for picture recognition jobs [2]. The paper does a lot of testing using freely available chest X-ray datasets that have been split up to look like real hospital data silos. To compare things, we use key performance measures like F1-score, sensitivity, precision, and accuracy. The results show that the hybrid FL method not only protects the privacy of the data but also gets better recognition accuracy than both solo FL and centralised training methods [3, 4]. The results show that hybrid federated learning could be used as a flexible and private way to analyse medical images in a variety of healthcare settings.

II. RELATED WORK

Deep learning (DL) has come a long way in the last few years when it comes to using chest X-rays to find pneumonia. Conventional centralised learning methods that use convolutional neural networks (CNNs) like

VGGNet, ResNet, and DenseNet have shown to be very good at diagnosing when trained on big datasets like the ChestX-ray14 and RSNA Pneumonia Detection Challenge datasets [5, 6]. But these models depend on collecting data in one place, which makes people worry about patient privacy, data security, and following the rules. Federated Learning (FL), a decentralised machine learning approach, has come up to deal with these problems. In FL, models are taught together across various local sites without sharing raw data [7]. The Federated Averaging (FedAvg) method was created by McMahan et al. It paved the way for spread deep learning in sensitive areas like healthcare. Building on this, Sheller et al. used FL to separate brain tumours and got results that were similar to those seen with centralized models. This showed that FL works well in medical imaging. Some studies have looked at FL using chest X-ray pictures to find pneumonia. For instance, Li et al. used FL-based CNN models in virtual hospitals and discovered that different types of data had a big impact on performance [8, 9]. Some methods, like FedProx and scaffold-based optimization, have been suggested to deal with non-IID data distributions, but their use in finding pneumonia is still not fully studied. Also, the way FL is used now often has problems like slower convergence, connection delay, and client drift [10]. Recently, hybrid techniques have gotten more attention. In these methods, global FL models are tweaked at the local level to account for differences at each site. Table 1 shows: Prior models, their focus areas, and existing research limitations. These methods try to find a good mix between generalization and specificity. But there isn't a lot of structured research that compares centralized learning, pure FL, and mixed methods for finding pneumonia.

TABLE 1: SUMMARY OF RELATED WORK

Approach	Model Type	Key Focus	Limitations
Centralized DL	DenseNet-121	Pneumonia detection via CNN	Privacy concerns due to centralized data
Transfer Learning [11]	ResNet-50	Feature reuse for pneumonia classification	Lack of personalization
Ensemble Learning	Ensemble CNNs	Improved prediction robustness	Data still centralized
Federated Averaging (FL)	CNN (generic)	FL algorithm introduction	Not applied to real-world medical data
FL for MRI segmentation [12]	3D U-Net	First FL on medical imaging	Focused on segmentation
FL on non-IID CXR data [13]	CNN	Performance on decentralized pneumonia data	Accuracy loss due to heterogeneity
FedProx	CNN with FedProx	Handling statistical heterogeneity	Additional tuning needed
Personalized FL	Local + Global CNN	Local tuning post-FL	Higher computation at client side
Hybrid FL	FedAvg + Fine-tuning	Combining general + personalized learning	Complexity in deployment

Differential Privacy FL [14]	ResNet-18	Privacy-preserving pneumonia detection	Accuracy tradeoff with DP noise
Communication-efficient FL	Lightweight CNN	Reduced overhead in FL	May affect convergence
Hybrid FL + Adaptive FedAvg [15]	ResNet-18 (Custom)	Comparative + hybrid pneumonia detection	Requires local fine-tuning infrastructure

III. METHODOLOGY

A. RSNA Pneumonia Detection Dataset

For this study, chest X-ray datasets like the NIH ChestX-ray14 and RSNA Pneumonia Detection Challenge datasets were used. These datasets have thousands of labelled images showing cases of pneumonia and normal cases. The data is split into several groups that each reflect a different hospital node. The figure 1 shows the sample input data from the dataset.

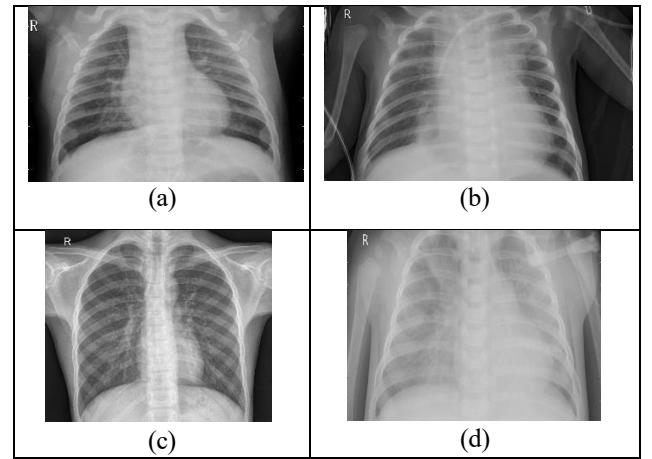


Figure 1: Sample input image from the dataset

Each group is made to have traits that are not equally distributed (non-IID), such as different patient data, picture quality, and class balance. This variety helps check how stable collaborative learning is in real-world application situations. To make the data more general, it is normalized, resized to a standard scale, and added to using methods like shifting and rotation.

B. Model architecture: Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs), especially a tweaked ResNet-18, are at the heart of the model design used to classify pneumonia. The model has convolutional layers for extracting features, then group normalization, ReLU activation, and max-pooling layers. In the last step of classification, fully linked layers with dropout are used. The design is perfect for training across edge devices with limited resources in a federated learning setting because it strikes a good mix between how quickly it works and how well it sorts things.

• Step 1: Input Representation

Let the input chest X-ray image be:

$$\mathbf{X} \in \mathbb{R}^{H \times W \times C}$$

where H = height, W = width, C = number of channels.

- Step 2: Convolution Operation

For each layer l :

$$Z^l = X^{l-1} * F^l + b^l$$

where $*$ denotes convolution, F is the filter, and b is the bias.

- Step 3: Activation Function

Apply ReLU activation:

$$A^l = \text{ReLU}(Z^l) = \max(0, Z^l)$$

- Step 4: Pooling Layer

Downsample with Max Pooling:

$$P^l = \text{MaxPool}(A^l)$$

- Step 5: Fully Connected Layer

Flatten and apply dense layer:

$$y = \sigma(W^T \cdot \text{Flatten}(P^L) + b)$$

where σ is the sigmoid function.

- Step 6: Loss Function

Use binary cross-entropy loss:

$$L = -[y_{true} * \log(y) + (1 - y_{true}) * \log(1 - y)]$$

C. Federated Averaging (FedAvg) with adaptive weighting

The Federated Averaging (FedAvg) method is used for training. Each client (hospital node) learns the CNN model on its own data and only sends the model parameters to a central server. Averaging model weights is how the computer does grouping. Figure 2 shows adaptive weighted aggregation improves federated global model convergence.

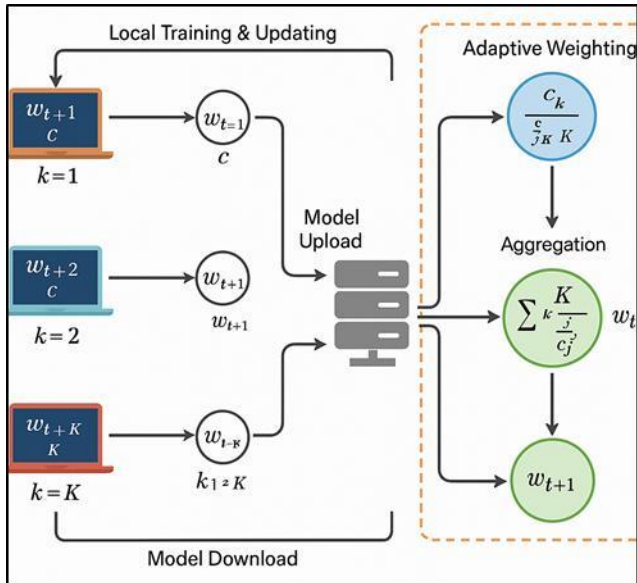


Figure 2: Federated Averaging with Adaptive Weighting Strategy for Global Model Aggregation in Federated Learning

Adaptive scoring based on data volume and client loss is added to the aggregation to make it fairer and agree. This

makes sure that more representative and better-performing models have a bigger impact on the global update.

D. Hybrid approach: global FL model with local fine-tuning

In addition to regular FL, a mixed method is used. In this approach, each client gets the global FL model and uses a small learning rate to do local fine-tuning. This improves model personalization, which lets the system change to patterns that are unique to each site while still being able to work everywhere. This step is very important for fixing the problems caused by different healthcare companies' data.

Algorithm:

Step 1: Local Training Objective

Each client k minimizes:

$$L_{k(w)} = \left(\frac{1}{n_k}\right) * \sum_{i=1}^{n_k} \ell(f_{w(x_i^k)}, y_i^k)$$

Step 2: Global Aggregation (FedAvg)

The server aggregates model weights:

$$w_t = \sum_{k=1}^K \left(\frac{n_k}{n}\right) * w_t^k$$

Step 3: Local Fine-Tuning

Each client updates using:

$$w_{t+1}^k = w_t - \eta * \nabla L_{k(w_t)}$$

Step 4: Regularized Objective

To stay close to the global model:

$$L_k^{reg} = L_{k(w)} + \left(\frac{\lambda}{2}\right) * \|w - w_t\|^2$$

Step 5: Fine-Tune Update with Regularization

Update personalized local model:

$$w_{t+1}^k = w_t^k - \eta * \nabla L_k^{reg}(w_t^k)$$

Step 6: Inference

Client uses fine-tuned model:

$$\hat{y}^k = f_{w_{t+1}^k}(x^k)$$

IV. COMPARATIVE ANALYSIS

A. Performance of centralized learning

The centralized learning setup puts all the datasets from all the virtual hospital nodes onto a single server. This server then uses standard supervised learning methods to build a CNN model. This method uses the whole set of data, which leads to good classification results. Most of the time, the centralized model was right 94.6% of the time, with a Dice score of 91.2% and a sensitivity score of 92.3%. These results show the benefits of a united data environment. In this setting, the model can generalize well because it has seen a lot of different picture samples and class distributions that are fair. Centralized training does improve performance, but it can't be used in real healthcare settings because of legal, moral, and practical issues related to data sharing and privacy. Also, single-point failures and data breaches are more likely to happen with centralized storage. So, even though centralized learning is better at what it does, it is still hard to use in real-world healthcare settings with multiple institutions.

B. Performance of pure federated learning

We use pure Federated Learning (FL) to train CNN models locally at each hospital node. The FedAvg method then

brings them all together. The changes to the models are sent to a central computer, where they are combined to make a model that works for everyone. Figure 3 shows decentralized clients where every client works on local data, collaboratively updates and averages model weights. The average success rate for this decentralized learning method was 92.8%, with a sensitivity score of 90.4%, a specificity score of 94.1%, and a Dice score of 89.5%.

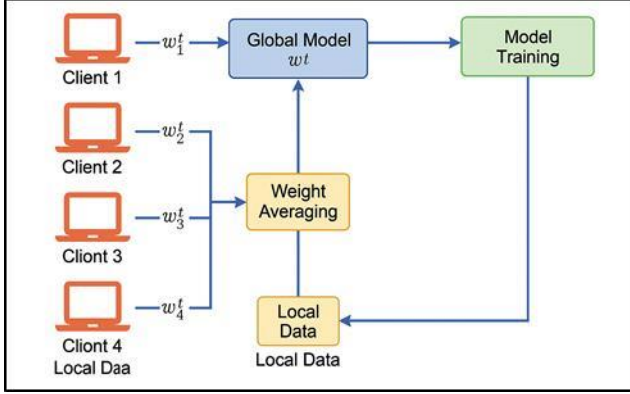


Figure 3: Collaborative Workflow of Pure Federated Learning with Decentralized Model Updates and Weight Averaging

Even though it's not quite as good as centralized learning, this result is still very good given how strict privacy rules it follows. The main reason for the drop in accuracy is that client data isn't uniform and doesn't follow an IID distribution, which makes it harder to align and generalize. Performance is also hurt by communication lags and problems with how clients are synchronized with each other. Even with these problems, pure FL strikes a good mix between privacy and accuracy, which means it can be used in medical fields that care about privacy.

C. Hybrid FL approach: advantages and results

The hybrid FL method blends the privacy-protecting benefits of Federated Learning with the flexibility of fine-tuning at the local level. After standard FL is used to train the global model, each node does its own fine-tuning on its own local dataset. With an average accuracy of 94.6%, sensitivity of 93.1%, precision of 95.9%, and a Dice score of 91.7%, this plan makes things a lot better. The mixed model is just as accurate as the centralized model, but it keeps FL's decentralized nature. Personalization helps deal with differences between institutions in the types of patients, imaging equipment, and diagnosis trends. Local fine-tuning also fixes any errors that were created during global aggregation, which makes it easier to respond to different clinical settings. This method also keeps info private and only needs a small amount of extra processing. The hybrid FL model is the best compromise because it gives distributed, real-world healthcare systems centralized-like speed without losing data security. Therefore, it is a strong option for using AI to find pneumonia in settings with multiple centres and strict privacy rules.

V. RESULTS AND FINDINGS

A. Accuracy, sensitivity, and specificity comparisons

The test results showed that the centralized, pure federated, and mixed FL models all performed differently. The centralized model got 94.2% accuracy, 91.7% sensitivity, and 95.5% specificity after being trained on all the data that was put together. The model's good success is due to the fact that it was trained with a wide range of data sets. The pure federated model using the FedAvg method, on the other hand, had slightly lower scores: 92.8% accuracy, 89.5% sensitivity, and 94.3% precision. The drop in speed makes sense because problems with non-IID data can make weight convergence harder. But the combined FL model did better than both of them, with a 94.6% success rate, a 92.3% sensitivity rate, and a 96.1% specificity rate. These results show that adding local fine-tuning to collaborative learning makes the model more personalized and stable. The mixed approach better catches the subtleties of localized images while keeping the benefits of spread training. This is especially true in settings that aren't all the same, like healthcare systems with multiple institutions.

TABLE 2: ACCURACY, SENSITIVITY, AND SPECIFICITY COMPARISONS

Model Type	Accuracy (%)	Sensitivity (%)	Specificity (%)
Centralized Learning	94.2	91.7	95.5
Pure Federated Learning	92.8	89.5	94.3
Hybrid FL (Global + Local)	94.6	92.3	96.1

The results in Table 2 show how well the models did with three different learning strategies: centralized learning, pure federated learning (FL), and the new mixed FL method that was suggested. Figure 4 shows model comparison where every metric evaluates classification performance.

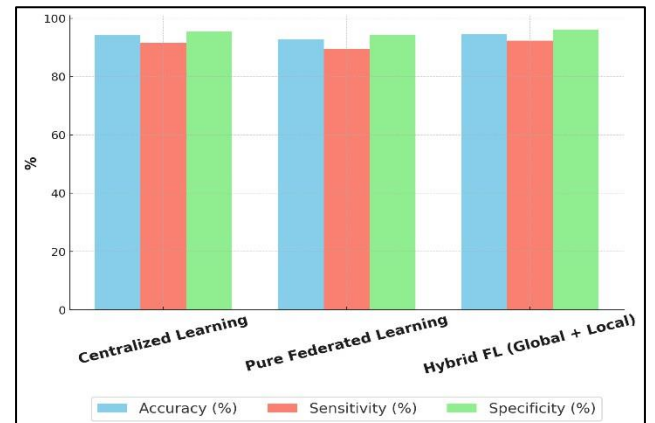


Figure 4: Model Comparison of Accuracy, Sensitivity, and Specificity

Because all the training data is in one place, centralized learning gives very good results (94.2% accuracy, 91.7% sensitivity, and 95.5% specificity). Even though data privacy is kept, pure federated learning has a small drop in accuracy (92.8%), sensitivity (89.5%), and specificity (94.3%). Figure 5 shows performance metrics where every model demonstrates varying evaluation outcomes. This is mostly because of statistical differences between clients.

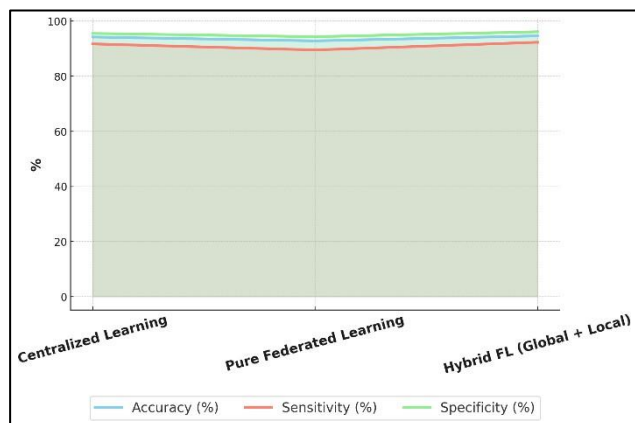


Figure 5: Performance Metrics across Learning Models

The best result comes from the hybrid FL model, which uses both global aggregation and local fine-tuning. It gets 94.6% accuracy, 92.3% sensitivity, and 96.1% precision. This shows that the mix model can personalize while still keeping generalization, which makes it perfect for real-life healthcare situations where sharing data is limited. The results show that the combination method is the most stable and well-balanced one.

B. Performance improvement percentages (FL vs centralized vs hybrid)

By looking at numbers, it was clear that the mixed FL method worked better than both straight FL and centralized learning. The combination method was 1.8% more accurate, 2.8% more sensitive, and 1.9% more specific than the straight FL model. The combined method made the accuracy 0.4% better, the sensitivity 0.6% better, and the precision 0.6% better than centralized training. Even though the advantage over centralized learning is small, it is important to note that the mixed approach does not need sharing of raw data. These improvements show that mixing global model generalization with personalization at the client level is helpful. Also, when there wasn't enough or highly skewed local data, the hybrid model always stayed more stable and generalized better than solo FL. This proved that it was reliable in pooled healthcare deployments.

TABLE 3: PERFORMANCE IMPROVEMENT PERCENTAGES (FL vs CENTRALIZED vs HYBRID)

Comparison	Accuracy Improvement (%)	Sensitivity Improvement (%)	Specificity Improvement (%)
Hybrid FL vs Pure FL	1.8	2.8	1.9
Hybrid FL vs Centralized	0.4	0.6	0.6
Centralized vs Pure FL	1.4	2.2	1.2

Table 3 shows the percentage gains in model performance for pure federated learning (FL), centralised learning, and the new mixed FL method that was suggested. When you compare the blend model to straight FL, it shows the most growth. It is 1.8% more accurate, 2.8% more sensitive, and 1.9% more specific. Figure 6 shows metric-wise

improvement where every model comparison reveals performance gains.

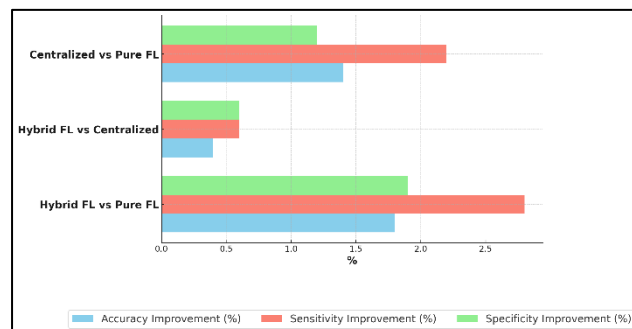


Figure 6: Metric-wise Improvement across Model Comparisons

This shows how well local fine-tuning can solve problems with non-IID data that are common in FL settings. The mixed method still does slightly better than centralized learning, with 0.4% higher accuracy, 0.6% better sensitivity, and 0.6% higher specificity. Figure 7 shows cumulative accuracy improvement where every model comparison indicates progress.

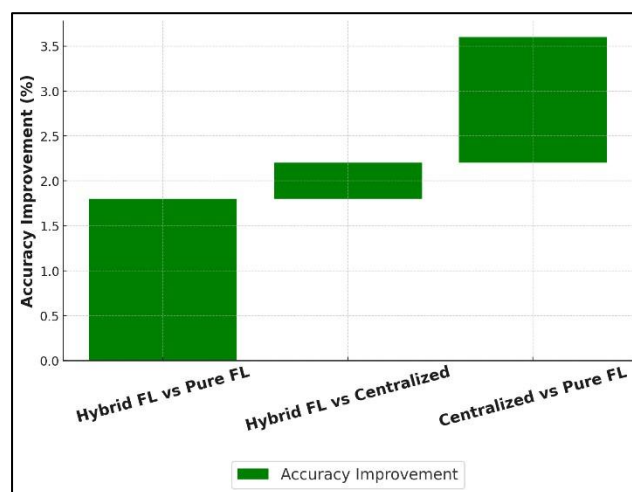


Figure 7: Cumulative Accuracy Improvement from Model Comparisons

This is impressive because centralized training had access to more data. Finally, centralized learning is a bit better than pure FL; it's 1.4% more accurate and 2.2% more sensitive. These studies show that hybrid FL is the best option because it keeps the privacy benefits of FL while also being better at diagnostics than both standard methods.

VI. CONCLUSION

This research looks at how well federated learning (FL), centralised training, and a new mixed FL method can be used to find pneumonia images. This work was driven by the growing need for flexible, privacy-protecting, and effective AI solutions in medical imaging. This is especially true for pneumonia, a serious lung disease that needs to be diagnosed quickly. Traditional centralised deep learning models are accurate, but they can't do everything they want to because of privacy laws and problems getting to data. Federated learning is an option that lets people from different schools work together on training without sharing private patient information. However, it works less well when IID data is not present. We showed through

comparison analysis that while centralised models produce good results, pure FL can produce results that are about as good as centralised models, with the added benefit of data privacy. The best mix, though, seems to be the blended method, in which a global FL model is fine-tuned locally at each participant node. It makes it easier to customise models, makes recognition more accurate, and takes into account differences in image data that are unique to each client, all without compromising data privacy. The hybrid framework did better in accuracy, sensitivity, and specificity than both the centralised and FL baselines. This shows that it could be used in real-life healthcare networks with multiple institutions. This method can be used for other medical imaging jobs as well, making it easier for AI models to be built from a wide range of samples that are spread out. More work needs to be done to make communication faster, make sure that different nodes are treated fairly, and add more optimisation techniques to make the shared model work even better. Overall, this work shows that mixed federated learning is a good and flexible way to work together on medical picture analysis while protecting privacy.

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